# #1. Tạo DataFrame

```
# ========================================
# #1. TAO DATAFRAME
# ========================================

import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Ho_va_ten": [
        "Nguyen Van An",
        "Tran Thi Binh",
        "Le Hoang Nam",
        "Pham Minh Anh",
        "Hoang Van Hung",
        "Vo Thi Lan",
        "Dang Quoc Bao",
        "Bui Ngoc Mai",
        "Do Thanh Cong",
        "Nguyen Thi Hoa"
    ],
    "Chuyen_can": [9, 8, 10, 7, 8, 9, 7, 10, 8, 9],
    "Giua_ky": [8, 7, 9, 6, 7, 8, 6, 9, 7, 8],
    "Cuoi_ky": [8.5, 7.5, 9, 6.5, 8, 8.5, 7, 9.5, 7.5, 8]
}

df = pd.DataFrame(data)

# Hien thi bang du lieu
display(df)
```

| | Ho_va_ten | Chuyen_can | Giua_ky | Cuoi_ky |
|---|---|---|---|---|
| 0 | Nguyen Van An | 9 | 8 | 8.5 |
| 1 | Tran Thi Binh | 8 | 7 | 7.5 |
| 2 | Le Hoang Nam | 10 | 9 | 9.0 |
| 3 | Pham Minh Anh | 7 | 6 | 6.5 |
| 4 | Hoang Van Hung | 8 | 7 | 8.0 |
| 5 | Vo Thi Lan | 9 | 8 | 8.5 |
| 6 | Dang Quoc Bao | 7 | 6 | 7.0 |
| 7 | Bui Ngoc Mai | 10 | 9 | 9.5 |
| 8 | Do Thanh Cong | 8 | 7 | 7.5 |
| 9 | Nguyen Thi Hoa | 9 | 8 | 8.0 |

# #2. Tính điểm tổng kết

```
# #2. TINH DIEM TONG KET
# ========================================

df["Tong_ket"] = (
    df["Chuyen_can"] * 0.2
    + df["Giua_ky"] * 0.3
    + df["Cuoi_ky"] * 0.5
).round(2)

display(df)
```

| | Ho_va_ten | Chuyen_can | Giua_ky | Cuoi_ky | Tong_ket |
|---|---|---|---|---|---|
| 0 | Nguyen Van An | 9 | 8 | 8.5 | 8.45 |
| 1 | Tran Thi Binh | 8 | 7 | 7.5 | 7.45 |
| 2 | Le Hoang Nam | 10 | 9 | 9.0 | 9.20 |


| | | | | | |
|---|---|---|---|---|---|
| 2 | Le Hoang Nam | 10 | 9 | 9.0 | 9.20 |

| 3 | Pham Minh Anh | 7 | 6 | 6.5 | 6.45 |
|---|---|---|---|---|---|
| 4 | Hoang Van Hung | 8 | 7 | 8.0 | 7.70 |
| 5 | Vo Thi Lan | 9 | 8 | 8.5 | 8.45 |
| 6 | Dang Quoc Bao | 7 | 6 | 7.0 | 6.70 |
| 7 | Bui Ngoc Mai | 10 | 9 | 9.5 | 9.45 |
| 8 | Do Thanh Cong | 8 | 7 | 7.5 | 7.45 |
| 9 | Nguyen Thi Hoa | 9 | 8 | 8.0 | 8.20 |

## *#3. Xếp loại*

```python
# #3. XEP LOAI SINH VIEN
# ========================================

def xep_loai(diem):
    if diem >= 8.5:
        return "Gioi"
    elif diem >= 7.0:
        return "Kha"
    elif diem >= 5.0:
        return "Trung binh"
    else:
        return "Yeu"

df["Xep_loai"] = df["Tong_ket"].apply(xep_loai)

display(df)
```

| | Ho_va_ten | Chuyen_can | Giua_ky | Cuoi_ky | Tong_ket | Xep_loai |
|---|---|---|---|---|---|---|
| 0 | Nguyen Van An | 9 | 8 | 8.5 | 8.45 | Kha |
| 1 | Tran Thi Binh | 8 | 7 | 7.5 | 7.45 | Kha |
| 2 | Le Hoang Nam | 10 | 9 | 9.0 | 9.20 | Gioi |
| 3 | Pham Minh Anh | 7 | 6 | 6.5 | 6.45 | Trung binh |
| 4 | Hoang Van Hung | 8 | 7 | 8.0 | 7.70 | Kha |
| 5 | Vo Thi Lan | 9 | 8 | 8.5 | 8.45 | Kha |
| 6 | Dang Quoc Bao | 7 | 6 | 7.0 | 6.70 | Trung binh |
| 7 | Bui Ngoc Mai | 10 | 9 | 9.5 | 9.45 | Gioi |
| 8 | Do Thanh Cong | 8 | 7 | 7.5 | 7.45 | Kha |
| 9 | Nguyen Thi Hoa | 9 | 8 | 8.0 | 8.20 | Kha |

# #4. Thống kê

```python
# ========================================
# #4. THONG KE
# ========================================

diem_trung_binh = df["Tong_ket"].mean()

sv_cao_nhat = df.loc[df["Tong_ket"].idxmax()]
sv_thap_nhat = df.loc[df["Tong_ket"].idxmin()]

so_sv_dat = (df["Tong_ket"] >= 5).sum()

print(f"Diem trung binh lop: {diem_trung_binh:.2f}")

print(
    f"Sinh vien cao nhat: {sv_cao_nhat['Ho_va_ten']} "
    f"- {sv_cao_nhat['Tong_ket']:.2f}"
)
```

```
)

print(
    f"Sinh vien thap nhat: {sv_thap_nhat['Ho_va_ten']} "
    f"- {sv_thap_nhat['Tong_ket']:.2f}"
)

print(f"So sinh vien dat: {so_sv_dat}/10")
```

```
Diem trung binh lop: 7.95
Sinh vien cao nhat: Bui Ngoc Mai - 9.45
Sinh vien thap nhat: Pham Minh Anh - 6.45
So sinh vien dat: 10/10
```

# #5. Trực quan hóa.

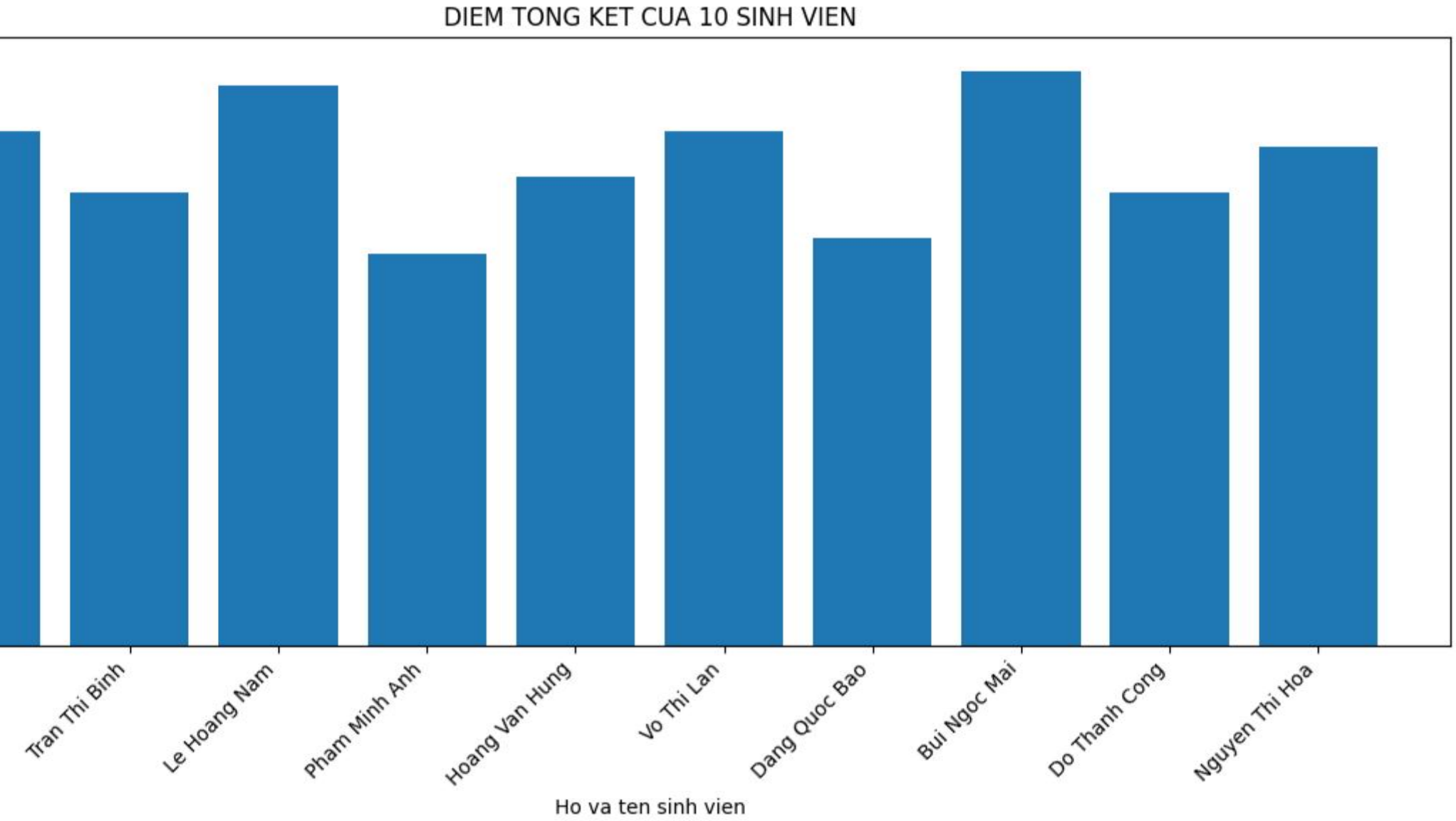
```
# ========================================
# #5. TRUC QUAN HOA DU LIEU
# ========================================

plt.figure(figsize=(12, 6))

plt.bar(df["Ho_va_ten"], df["Tong_ket"])

plt.title("DIEM TONG KET CUA 10 SINH VIEN")
plt.xlabel("Ho va ten sinh vien")
plt.ylabel("Diem tong ket")
plt.ylim(0, 10)
plt.xticks(rotation=45, ha="right")

plt.tight_layout()
plt.show()
```


DIEM TONG KET CUA 10 SINH VIEN
Tran Thi Binh
Le Hoang Nam
Pham Minh Anh
Hoang Van Hung
Vo Thi Lan
Dang Quoc Bao
Bui Ngoc Mai
Do Thanh Cong
Nguyen Thi Hoa
Ho va ten sinh vien